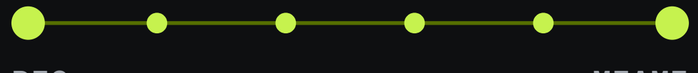
TESTNET · TEST KEYS · NOT REAL CUSTODIANS

Radox Custody Rail

Know who holds the goods: every handover signed by both custodians.

Toward a SWIFT-like custody standard on Stellar · Pilot corridor: Rio–Miami

Scan: the live shipment
rio-miami-rail.vercel.app



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PMF · THE PROBLEM

Who holds the cargo right now?

Between seller and buyer, a shipment changes hands several times.



Handovers are mostly paper, or sit inside each operator's own system: rarely one shared record across borders.

Our hypothesis: a financier cannot see who holds the goods before paying. Mentors were supportive; informal feedback, no customer has confirmed the need yet.

US$2.5T

Global trade finance gap.

ADB Global Trade Finance Gap Survey, Dec 2025. It sizes the financing gap; it is not evidence of the custody problem.

Context: our explainer video about a Rio–Miami trade-corridor initiative poses one possibility: a unique digital record for every shipment that both sides can check. This project is a custody layer for that record.

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THE SOLUTION

Both custodians sign, or nothing changes

Outgoing custodian
`signs`

One Stellar transaction
`custody_registry.handover`

Incoming custodian
`signs`

Either signature missing → the call fails.

— Each handover moves custody and extends an on-chain hash chain, the `head`.

— The token stays with its holder in transit; only custody changes.

— After Delivered, the operator burns the token with the existing `redeem`[1].

A custody record keyed to the container number. In production it would sit beside the carrier's bill of lading; not integrated in this build.

[1] Radox platform policy, not enforced on-chain by the registry; redeem still needs the holder's signature.

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THE PRODUCT · LIVE ON TESTNET

Anyone can check the chain. No login.

Warehouse → Port, both signed	ledger 5381142 · 635c99…f0bd38
Port → Carrier, both signed	ledger 5381143 · 56a0f0…7b7cc2
Event attested by Source	ledger 5381144 · 333792…068905
Carrier → Customs, both signed	ledger 5381145 · ac268a…c3c4b2
Customs → Recipient, both signed	ledger 5381146 · 5199d4…400b7a
Delivered by Recipient to Holder, both signed	ledger 5381147 · 919def…144e37
Redeemed: token burned	864fd1…15644d

Verify head

Verify head recomputes the hash chain in the browser and compares it with the head stored in the contract.

MATCH

Clicked live on 25 Oct 2026.

Read-only: Stellar RPC and Horizon, no wallet.

rio-miami-rail.vercel.app

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TECHNICAL WORKFLOW

7 testnet transactions. All successful.

open	handover 1	handover 2	handover 3	handover 4	deliver	burn
operator	Warehouse → Port	Port → Carrier	Carrier → Customs	Customs → Recipient	Recipient → Holder	existing redeem
2b3d51f0	635c99ac	56a0f006	ac268a49	5199d4fb	919def8c	864fd164

7
successful testnet transactions

9
invalid calls refused (simulated)

24
contract unit tests pass

MATCH
verify-head rebuilds the chain

`custody_registry` CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII

Plus an attestation (33379211); full hashes in proofs/lifecycle.json. A second run (branch three-machine, proofs/three-machine.json): two team members authorised handovers on their own computers, with their own testnet keys.

All are test keys, most held by one harness: this proves the mechanism, not independent custodians.

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ECOSYSTEM · WHY STELLAR

Why Stellar: two signers, one call

01	**Multi-party auth in one call**	Soroban authorises both custodians inside one contract invocation: the handover is atomic.
02	**Events plus an on-chain head**	RPC keeps contract events about 7 days. The head stored in the contract lets anyone rebuild the full history later.
03	**The token stays put, then burns**	The shipment token never moves in transit; the existing Radox redeem burns it after delivery.

Nearest neighbours we found on Stellar (LumenLoop, SCF, Scout, 24–25 Oct): Kutana and Trustless Work release escrow on trade milestones; Pipeline finances commodity trade. We found none that records a custody handover signed by two custodians, and no SEP.

Open source, Apache-2.0: a custody primitive other Stellar real-world-asset issuers can reuse.

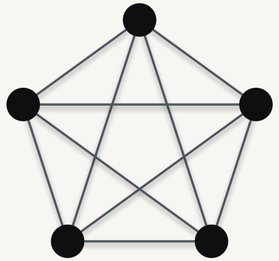
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USERS · BUSINESS MODEL

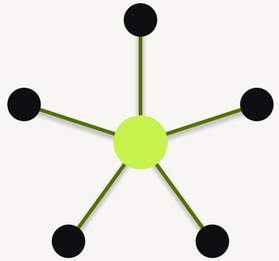
Integrate once, reach every party that joins

With one public custody standard, a port, airport, carrier, chamber or financier would integrate once and reach every other party: adopt without a custom project.

PAIR BY PAIR · 10 LINKS



ONE RADOX RAIL · 5 LINKS



Example, 5 parties: 10 custom links, or 5 links to one standard.

OPEN · ANYONE

— Record format, roles and events: the custody_registry contract (Apache-2.0)

— Verify any shipment, no login (slide 4)

RADOX OPERATES · CLIENT PAYS

— Custodian onboarding, issuance controls, operation

— The API and backend: closed, not self-hostable

— Paid by the financial institution; not priced

Adoption still takes work: a key per party, its own signatures, one connection to the rail (the Radox API).
Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

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HACKATHON PROGRESS

Built this weekend, on disclosed prior work

PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform (closed backend); its 3 Soroban contracts are on Stellar mainnet, public since 2026-08-19

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Architecture notes (20 Oct), planning files (24–25 Oct), the explainer video, our agent tooling (from 22 Oct)

— No contract code before then: first repo commit 25 Oct 09:00 UTC

BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script, the public shipment page, this deck

— Radox API reads the registry: same head (closed backend, testnet)

AI attribution Code, tests, scripts, the shipment page, docs and this deck were made with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

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VISION · NEXT · THE ASK

Vision: one custody standard, every corridor

Rio–Miami is our pilot corridor. The goal is a public, SWIFT-like custody standard on Stellar for tokenized goods in transit worldwide, to integrate with the systems of chambers of commerce, ports and airports.

NOW

Mechanism proven on testnet. No standard, users or revenue yet.

NEXT

One Rio–Miami pilot shipment (none agreed yet). Full test matrix.

PLANNED

SCF Build, then a security audit (neither done). Mainnet.

FUTURE

More corridors; port, airport, chamber links (none confirmed); USD leg (not built).

Our ask **Introductions to one financial institution and one port, airport or logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

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THE TEAM

Why us: a founder already on Stellar mainnet

Pedro Wakigawa Saragossy

Founder, captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

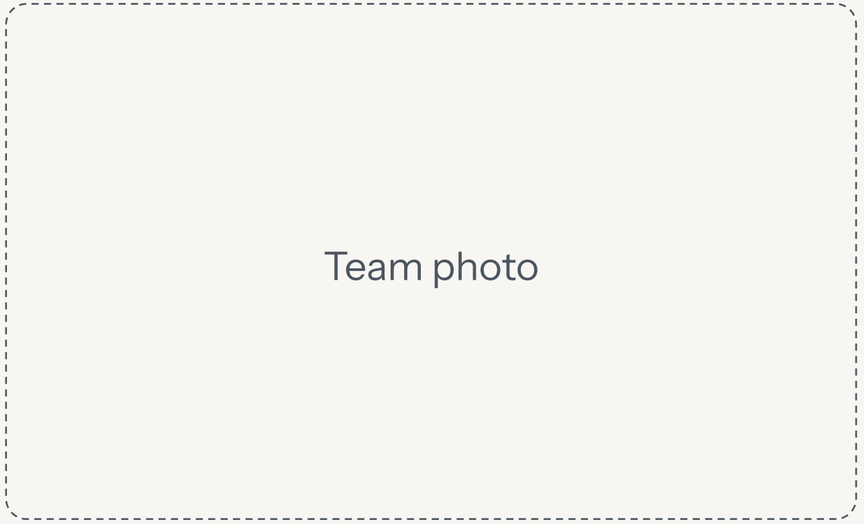
Leo Schlanger

Developer: custody_registry, the testnet harness and the proofs.

Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.



Rio–Miami first. Our goal: one custody standard for every corridor.

Live page rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

AI Code, page, docs and deck made with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

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